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Construction of core-shell structured SiO_2@MoS_2 nanospheres for broadband electromagnetic wave absorption

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ABSTRACT

Herein, core-shell structured SiO_2@MoS_2 nanospheres were made via hydrothermal method, with MoS_2 nanosheets growing irregularly on the outside surface of SiO_2. The microstructure and EM wave absorption properties of the prepared SiO_2@MoS_2 nanospheres were explored. SiO_2@MoS_2 achieved a minimum reflection loss (RL_{min}) value of − 49.35 dB with a thickness of 2.8 mm, and an effective absorption bandwidth (EAB) of 7.44 GHz. The remarkable EM wave absorption performance of SiO_2@MoS_2 nanospheres could be attributed to interfacial polarization originating from the designed core-shell structure and the multiple reflections on MoS_2 ultrathin nanosheets. Meanwhile, impedance matching was regulated effectively by combination of SiO_2 and MoS_2, which provided a synergistic effect and enhanced EM wave dissipation. Therefore, the core-shell structured SiO_2@MoS_2 could be a great potential candidate material in the application of high-performance EM wave absorbing area.

1. Introduction

The quick advancement of electronic information technology, especially the widespread use of electronic devices, has caused increasingly serious EM pollution [1–3]. Currently, EM pollution has become one of the main pollutions in the 21st century, posing a significant threat to both human health and the natural environment. To eliminate the menace of EM pollution, EM wave absorbing materials with "wide effective absorption bandwidth, strong reflection loss, light weight and thin thickness" should be pursued [4,5].

MoS_2 has received a lot of interest as a potential EM wave absorber because of its abundant defects and high specific surface area, causing strong dielectric loss [6]. MoS_2 with 2D lamellar structure is constrained by weak Van Der Waals forces and it has 2H and 1T two phases [7–9]. Currently, a good deal of studies for the EM wave absorbing properties of MoS_2-based composites are mainly focused on the 2H phase. For example, Luo et al. synthesized ZnO/MoS_2 hybrids and reached the optimal reflection loss of − 35.80 dB [10]. Fe@MoS_2 composites with a porous coin-like structure were effectively manufactured by Pan et al., and the RL_{min} value was determined to be − 37.00 dB with the EAB was 4.73 GHz, which was attributed to the cooperative effect of magnetic loss and dielectric loss [11]. MXene-MoS_2 hybrids were designed by Li et al., and the RL_{min} value of − 46.72 dB could be achieved. The results pointed out that the formation of 3D conductive interconnected network and the synergy of the 2D heterostructures can elevate the dielectric loss ability of the materials [12]. However, the optimization and improvement of EM wave absorption for 2H-MoS_2 is confined due to its poor conductivity [13]. While 1T-MoS_2 exhibits good conductivity, mixed-phase MoS_2 can be employed to optimize EM parameters by improving dielectric loss [14].

Additionally, MoS_2 nanosheets have agglomerate propensity because of the Van Der Waals and high surface energy in layers, which leads to poor impedance matching [15]. To avoid this drawback, many studies have proved that structural design was an efficient way to improve the restricted EM wave absorption capability of MoS_2, including porous structure [16], core-shell structure [17], hollow structure [18] and flower-like structure [19]. In particular, the construction of core-shell structure is beneficial for the optimization of impedance matching combining with other materials and the generated interfaces can increase the dielectric loss performance of MoS_2 composites [20]. Wang et al. synthesized MoS_2@Fe_3O_4-GNs composites with hollow spherical structures, obtaining satisfactory performance with a RL_{min} of −48.10

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dB [21]. Zhang et al. successfully fabricated core-shell CS/MoS_2 composites and achieved a RL_{min} value of − 52.60 dB with the thickness was 1.4 mm. They pointed out that the primary reasons for the high absorbing performance were the reflected EM wave and the formed interfaces [22]. Thus, the forming core-shell structure can effectively inhibit the aggregation of MoS_2 nanosheets and increase EM wave absorption performance by improving interfacial polarization. SiO_2 as excellent wave-transparent material is a desired candidate to adjust EM parameters and improve impedance matching [23,24]. For example, Bai et al. [25] prepared Fe_3O_4 microspheres coated with SiO_2 (SiO_2@-Fe_3O_4), which could effectively enhance the dispersion of Fe_3O_4 and optimize EM parameters. The RL_{min} value of SiO_2@Fe_3O_4 was reduced from −35.7 dB to −43.6 dB and the EAB was expanded from 3.60 GHz to 7.00 GHz. Guo et al. [26] constructed N-doped Ni@SiO_2/graphene composites and found that low-dielectric SiO_2 could improve impedance matching and attenuation capability by regulating the coating thickness, reaching the RL_{min} value of −71.13 dB (2.46 mm). However, construction of core-shell structured SiO_2@MoS_2 composites have not been considered, therefore it remains space to investigate and improve the EM wave absorption performance of MoS_2 nanosheets by introducing SiO_2.

In this work, core-shell structured SiO_2@MoS_2 nanospheres were successfully prepared via a simple hydrothermal process. By varying the MoS_2 contents, the performance of EM wave absorption was investigated. The results indicated that SiO_2@MoS_2 exhibited outstanding EM wave absorption (RL_{min} of − 49.35 dB), corresponding to EAB of 7.00 GHz at 2.8 mm, and an EAB_{max} of 7.44 GHz at 3.2 mm. The synergy of rich polarization loss, as well as proper impedance matching contributes to broadband EM wave absorption of core-shell structured SiO_2@MoS_2 nanosphere.

2. Experimental sections

2.1. Materials

Thioacetamide (C_2H_5NS, ≥ 99.7%) and sodium molybdate ($Na_2MoO_4{\cdot}2H_2O$, ≥ 99.7%) were acquired from Tianjin Zhiyuan Chemical Reagent Co., Ltd. and Tianjin Yongda Chemical Reagent Co., respectively. Ammonia (analytical purity) was provided by Guangdong Xilong Technology Co., Ltd., while ethylorthosilicate (TEOS) was supplied by Tianjin Kemer Chemical Reagent Co., Ltd. Tianjin Fengchuan Co., Ltd. was supplied ethanol absolute (≥99.7%).

2.2. Preparation of SiO_2 microspheres

SiO_2 microspheres were synthesized by the Stöber method [27,28]. First, 40 mL of ethanol and 5 mL of $NH_3{\cdot}2H_2O$ were combined and thoroughly mixed by stirring for 20 min. The combined solution was then stirred at room temperature for 8 h while 2.8 mL of TEOS was added drop by drop. The white goods that had been made after cleaning with deionized (DI) water and drying for four hours.

2.3. Preparation of core-shell structured SiO_2@MoS_2

Core-shell structured SiO_2@MoS_2 nanospheres were obtained by a facial hydrothermal method [29,30]. 0.06 g of the prepared SiO_2 microspheres were dissolved in 10 mL of ethanol and ultrasonically treated for 20 min. 40 mL of DI water was combined with 2.90 g $Na_2MoO_4{\cdot}2H_2O$ and 3.60 g C_2H_5NS, and then the mixture was agitated for 30 min. The above two solutions were then combined and put into a 100 mL of Teflon-lined reactor autoclave for 20 h at 200 °C. The black precipitate was collected and washed three times with DI water and ethanol after cooling to room temperature. The products were obtained after drying, noted as SiO_2@MoS_2-2. SiO_2@MoS_2-1 and SiO_2@MoS_2-3 with different proportions were obtained by the same synthesis method, wherein the amounts of $Na_2MoO_4{\cdot}2H_2O$ and C_2H_5NS were 2.60 g and 3.30 g, 3.10 g and 3.90 g, respectively.

2.4. Characterization

X-ray diffraction (XRD) via Cu-Kα radiation (λ = 0.154 nm) was used to analyze the crystal structure of SiO_2@MoS_2 nanosphere. The morphology of SiO_2@MoS_2 nanospheres was examined using the scanning electron microscope (SEM, JSM-7001F), as well as the transmission electron microscope (TEM, JEOL-2100f). By using X-ray photoelectron spectroscopy (XPS, Thermo Fisher Scientific), the elemental composition and chemical states of SiO_2@MoS_2 nanospheres were examined. Raman spectroscopy of SiO_2@MoS_2 was recorded by a Thermofisher Dxr2xi with 523 nm excitation line of He-Ne laser. The electromagnetic parameters at 2.00–18.00 GHz were measured by vector network analyzers (MS46322B, Anli Co., Ltd, Japan). SiO_2@MoS_2 nanospheres (50%) and paraffin wax (50%) were evenly mixed and pressed into a ring shape (see Fig. 1).

3. Results and discussion

The XRD patterns of all the samples are presented in Fig. 2a. The broad diffraction locates between 15.6° and 30.7° indicates SiO_2 has an amorphous structure [31,32]. The diffraction peaks at 9.2° and 18.4° can be retrieved from (0 0 2) and (0 0 4) planes of 1T-MoS_2, while the peaks at 32.3° and 57.1°can be retrieved from the (1 0 0) and (1 1 0) planes of 2H-MoS_2 (JPCDS card no. 37-1492) [33,34]. The SiO_2@MoS_2 nanosphere exhibits the characteristic 1T-MoS_2 and 2H-MoS_2 diffraction peaks, proving that 1T/2H-MoS_2 was successfully synthesized. The diffraction peak of SiO_2 does not appear in the XRD patterns of SiO_2@MoS_2, because of the relatively weak intensity of diffraction peaks of SiO_2 [35,36]. To further analyze the structural information of SiO_2@MoS_2, Raman spectra of SiO_2@MoS_2 between 140 and 600 cm^{-1} is conducted, as shown in Fig. 2b. The characteristic peaks, which are seen at 287, 379, and 407 cm^{-1}, corresponding to the E_{1g}, E^1_{2g}, and A_{1g} Raman modes of 2H-MoS_2, respectively. Meanwhile, the SiO_2@MoS_2 samples show additional strong peaks at 150, 227, and 341 cm^{-1}, assigning to the J_1, J_2, and J_3 Raman modes of 1T-MoS_2, which is accordance with the XRD results.

The elemental composition and chemical states of SiO_2@MoS_2-2 are investigated by XPS. The SiO_2@MoS_2-2 contains O, Mo, S, and Si elements (Fig. 3a), confirming the successful combination of SiO_2 microspheres and MoS_2 nanosheets. The Si 2p peak with binding energy of 103.7 eV may be due to successful synthesis of SiO_2 (Fig. 3b) [37]. Fig. 3c shows the XPS spectra of Mo 3d. The peaks located at 228.8 eV and 231.9 eV correspond to Mo $3d_{5/2}$ and Mo $3d_{3/2}$ of 1T-MoS_2, while the Mo $3d_{5/2}$ peak (229.8 eV) and Mo $3d_{3/2}$ peak (233.1 eV) correspond to 2H-MoS_2 [38–40]. In addition, slightly weak peaks at 226.1 eV and 236.1 eV can be attributed to S 2 s and Mo^{6+}, respectively [41]. In Fig. 3d, S 2p spectra can be deconvoluted into S $2p_{3/2}$ (161.7 eV) and S $2p_{1/2}$ (162.9 eV), indicating the presence of S^{2-} in MoS_2 [42,43]. Fig. 4a reveals that pure MoS_2 is composed of a large number of nanosheets and polymerizes to form nanoflower structure. It can be observed that SiO_2 microspheres disperse uniformly with a size of 300 nm and a glossy outside surface (Fig. 4b). As for SiO_2@MoS_2 microspheres, MoS_2 nanosheets are irregularly anchored on the surface of SiO_2 microspheres, displaying undifferentiated morphology and structure (Fig. 4c-e). It is worth noting that SiO_2 microspheres are not obvious, which is caused by the encapsulation of MoS_2 nanosheets [44]. TEM and High-resolution TEM (HRTEM) images of representative SiO_2@MoS_2-2 are used to further understand the detailed morphology of SiO_2@MoS_2 microspheres (Fig. 4f and g). The ultra-thin MoS_2 nanosheets were wrapped on the spherical SiO_2, indicating successful preparation of core-shell structured SiO_2@MoS_2. The uniformly dispersed SiO_2@MoS_2 microspheres can provide a rich heterogeneous interface to enhance interface polarization. Additionally, ultra-thin MoS_2 nanosheets are conducive to enhancing multiple EM wave reflection and scattering [45,46]. It is

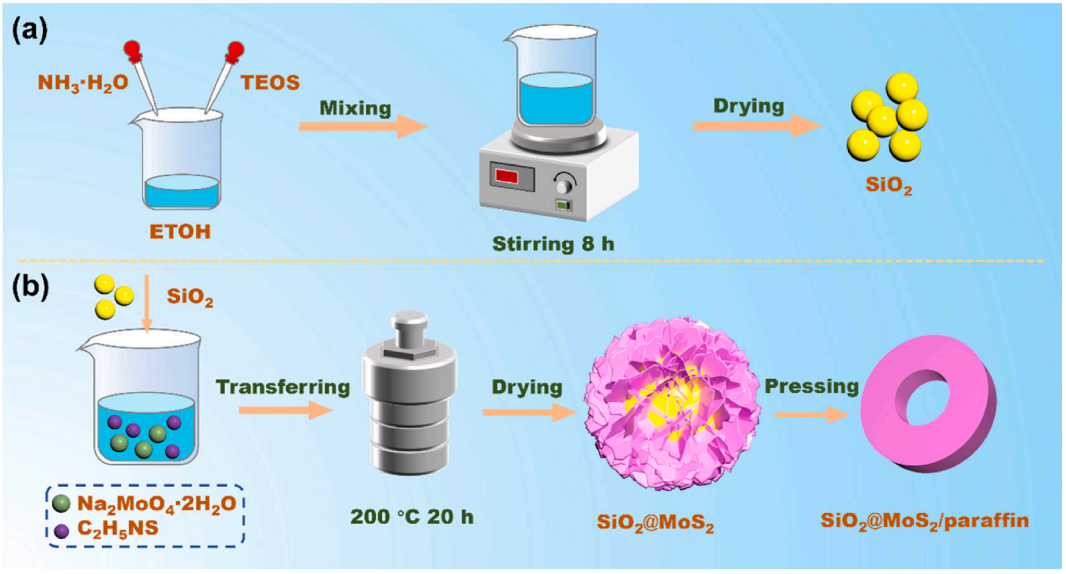


Fig. 1. The schematic of the preparation procedures of SiO_2@MoS_2.

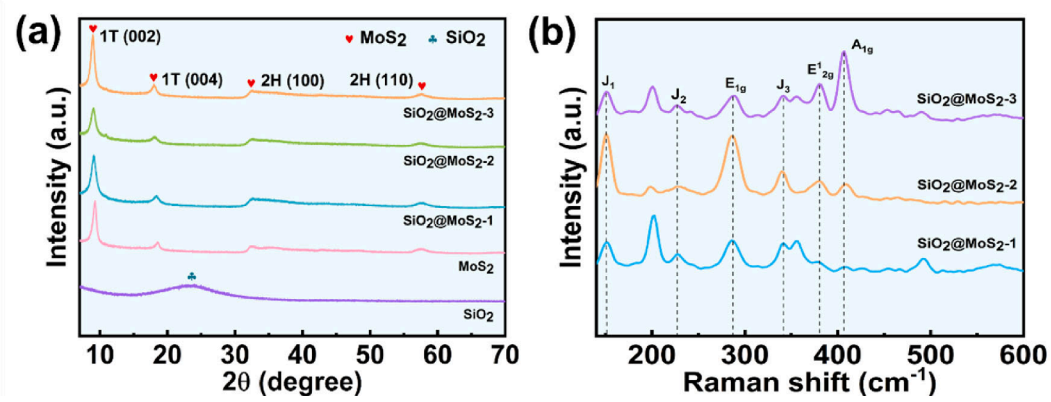


Fig. 2. (a) XRD patterns of SiO_2, MoS_2, and SiO_2@MoS_2 nanospheres. (b) Raman spectra of SiO_2@MoS_2 nanospheres.

found that the d-spacing of about 0.26 nm can be assigned to the (1 0 0) crystal plane of MoS_2 in the HRTEM image (Fig. 4h). Meanwhile, the element mapping images of SiO_2@MoS_2-2 confirm the uniform distribution of O, S, Si, and Mo elements (Fig. 4i).

EM parameters are significant indicators for the evaluation of the EM wave performance of materials. Because of the nonmagnetic nature of SiO_2 and MoS_2, the complex permittivity ($\varepsilon_r = \varepsilon' - j\varepsilon''$) of SiO_2@MoS_2 nanospheres is the only considered parameter. Generally speaking, the real part ε' reflects the ability to store energy, and the imaginary part ε'' indicates the ability to lose energy. The Debye theory gives the following expression for the ε' and ε'':

$$\varepsilon_r = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{1 + j2\pi f\tau} \tag{1}$$

$$\varepsilon' = \varepsilon_\infty + \frac{\varepsilon_s - \varepsilon_\infty}{1 + (2\pi f)^2\tau^2} \tag{2}$$

$$\varepsilon'' = \frac{2\pi f\tau(\varepsilon_s - \varepsilon_\infty)}{1 + (2\pi f)^2\tau^2} \tag{3}$$

where τ stands for the polarization relaxation time, ε_s represents the static permittivity and ε_∞ stands for the relative permittivity at high frequency. The ε' and ε'' of all materials have a significant downward trend in 2.00–18.00 GHz (Fig. 5a and b), showing a typical scattering behavior caused by relaxation polarization [47]. It is evident that the ε' and ε'' values of pure MoS_2 decline from 9.85 to 6.62 and 3.34 to 2.02, respectively, while SiO_2@MoS_2 exhibits larger ε' and ε'' values. The ε' values of SiO_2@MoS_2-1, SiO_2@MoS_2-2 and SiO_2@MoS_2-3 decline from 13.41 to 8.38, 5.37 to 6.07 and 12.34 to 8.15, demonstrating that proportion of MoS_2 in the SiO_2@MoS_2 can efficiently influence the EM parameters. Similarly, the ε'' values of the three samples are from 8.13 to 3.76, 6.91 to 2.85 and 5.53 to 2.92, respectively. In addition, the ε' and ε'' values of SiO_2@MoS_2-2 have obvious frequency-dependent fluctuations at 8–16 GHz, which could be caused by multiple dielectric relaxations [48,49], provoking the strongest interface polarization, conduction loss and dipole polarization [50,51]. The corresponding dielectric loss tangent ($\tan\delta_\varepsilon = \varepsilon''/\varepsilon'$) of all materials can be seen in Fig. 5c, where the curves have a similar tendency with the ε'' curves. After the introduction of the core-shell structure, the $\tan\delta_\varepsilon$ values of SiO_2@MoS_2 are much higher than those of pure MoS_2, which means that the dielectric loss capacity of SiO_2@MoS_2 is significantly improved. SiO_2@MoS_2-2 has the highest $\tan\delta_\varepsilon$ values, indicating it possesses the strongest EM wave dissipation capability.

Reflection loss (RL) is an intuitive metric to assess the ability to absorb EM waves of materials. The calculation equations are:

$$RL\& = 20\log\left|\frac{Z_{in} - Z_0}{Z_{in} + Z_0}\right| \tag{4}$$

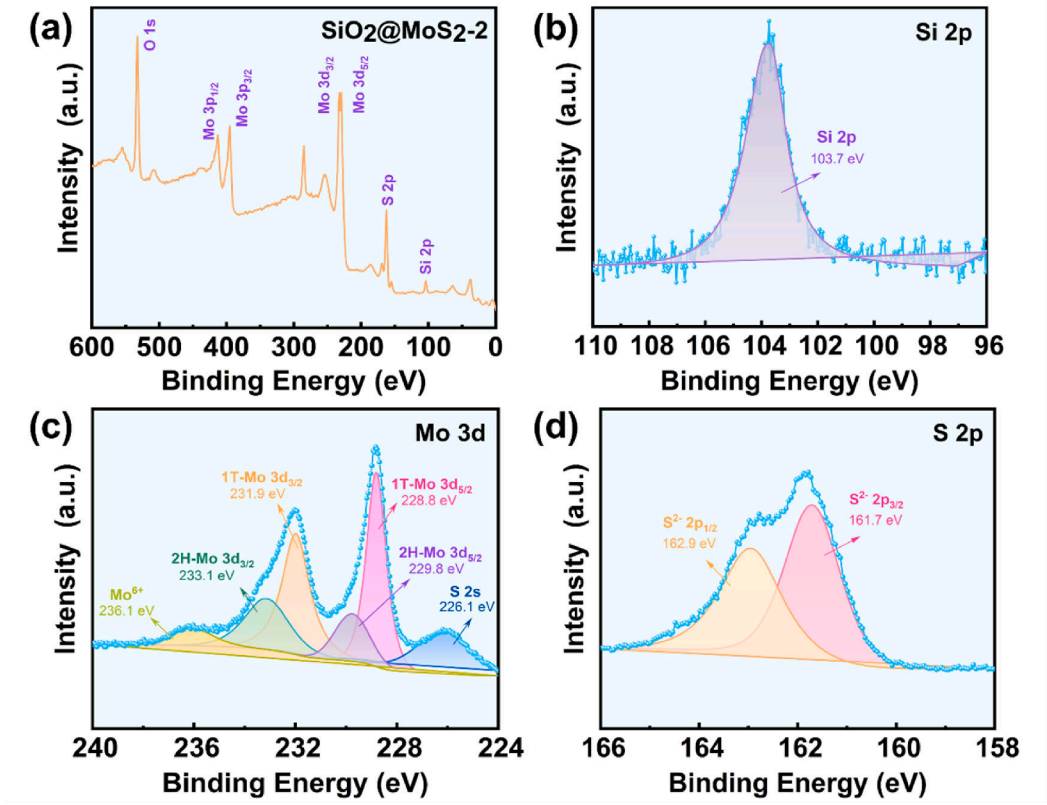


Fig. 3. (a) XPS survey; XPS spectra of (b) Si 2p; (c) Mo 3d; (d) S 2p of SiO_2@MoS_2-2 nanospheres.

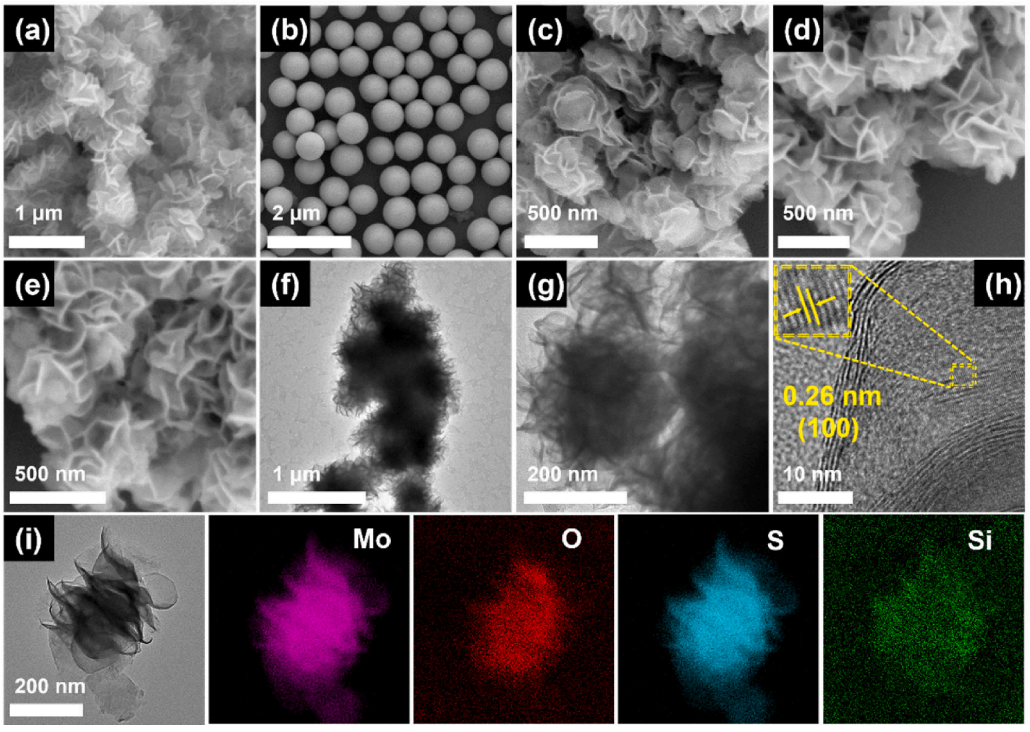


Fig. 4. SEM images of (a) pure MoS_2; (b) SiO_2; (c) SiO_2@MoS_2-1; (d) SiO_2@MoS_2-2; (e) SiO_2@MoS_2-3; (f, g) TEM images of SiO_2@MoS_2-2; (h) HRTEM images of SiO_2@MoS_2-2; (i) Elemental mapping images of SiO_2@MoS_2-2.

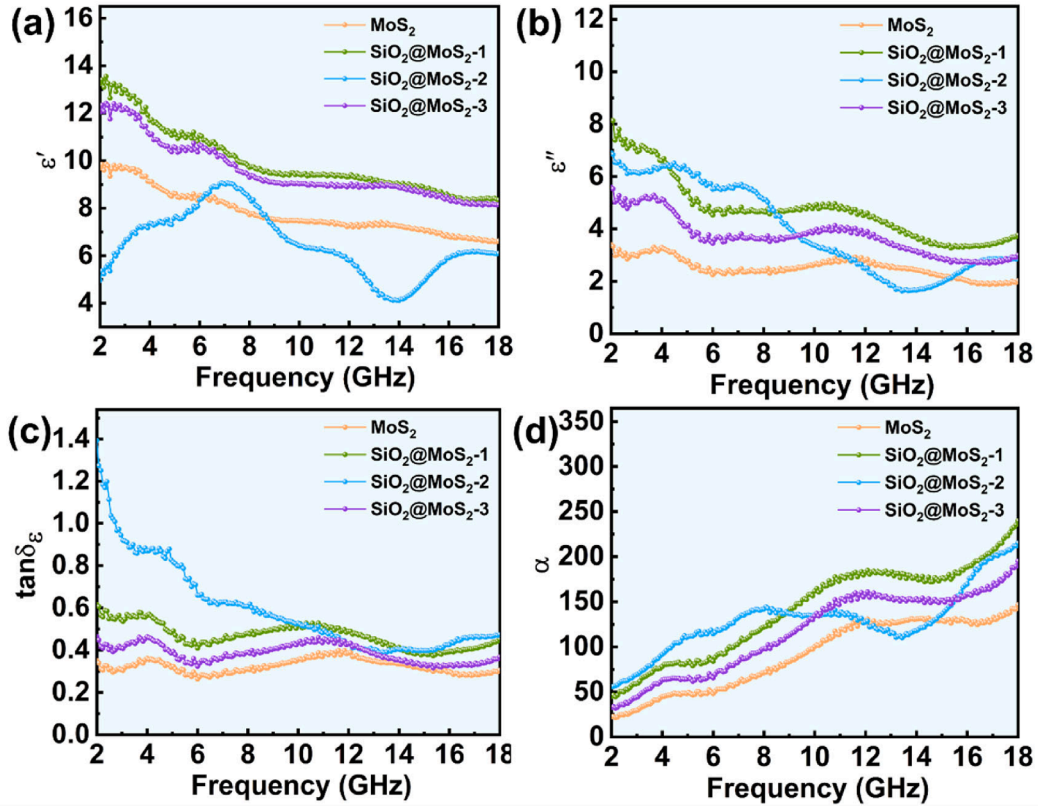


Fig. 5. (a) Real part of permittivity; (b) Imaginary part of permittivity; (c) Dielectric loss tangent; (d) Attenuation constant of SiO_2@MoS_2.

$$Z_{in} = Z_0\sqrt{\frac{\mu_r}{\varepsilon_r}}tanh\left(j\frac{2\pi fd}{c}\sqrt{\mu_r\varepsilon_r}\right) \quad (5)$$

Generally speaking, the absorbers can absorb and attenuate more than 90% of the EM wave when the RL value is less than − 10 dB [52–54]. Fig. 6 displays the RL maps of SiO_2@MoS_2 as the function of frequency (2.00–18.00 GHz) and thickness (1.0–5.0 mm). The RL_{min}

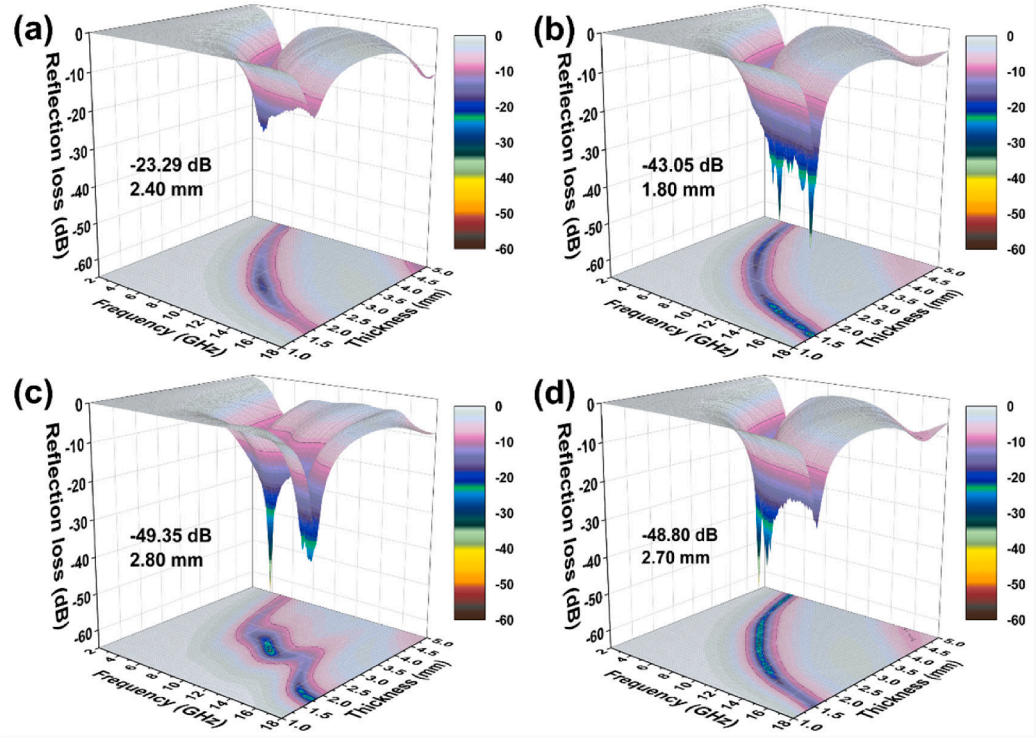


Fig. 6. 3D reflection loss curves and contour map of (a) MoS_2; (b) SiO_2@MoS_2-1; (c) SiO_2@MoS_2-2; (d) SiO_2@MoS_2-3.

value of pure MoS_2 is − 23.29 dB at 11.80 GHz and the corresponding thickness is 2.4 mm, exhibiting poor microwave attenuation properties. The microwave absorption ability of the absorbers is significantly enhanced when MoS_2 nanosheets grow erratically on SiO_2, which may be explained by the enhancement in conductivity loss and polarization loss. The RL_{min} values of the SiO_2@MoS_2 first increase and then decrease with the addition of MoS_2 loading, and SiO_2@MoS_2-2 exhibits the preferably EM wave absorption efficiency. By comparison, the RL_{min} of SiO_2@MoS_2-1 and SiO_2@MoS_2-3 are − 43.05 dB at 1.8 mm and − 48.80 dB at 2.7 mm (Fig. 6b and d), respectively. The RL_{min} value of SiO_2@-MoS_2-2 reaches − 49.35 dB with a thickness of 2.8 mm, and the largest EAB covers 7.44 GHz (6.88–14.32 GHz) at 3.2 mm (Fig. 6c). Furthermore, the effective EM wave absorption in almost all bands (including C, X, and Ku three bands) is achieved by adjusting the thickness of the SiO_2@MoS_2-2 (Fig. 7a). It is important to note that the RL_{min} value of all samples moves towards the lower frequency when the thickness grows. And the quarter-wavelength theory can explain this phenomenon [55]:

$$t_m = \frac{n\lambda}{4} = \frac{nc}{\left(4f_m\sqrt{|\mu_r||\varepsilon_r|}\right)n} = (1, 3, 5\cdots) \quad (6)$$

where f_m stands for matching frequency and t_m represents the thickness, respectively. Compared with the previously reported MoS_2-based absorbers, as shown in Fig. 7c [6,11,21,52,56–58]. Obviously, SiO_2@-MoS_2-2 exhibits outstanding microwave absorption efficiency and excellent effective absorption bandwidth.

Based on the efficient EM wave absorption ability of SiO_2@MoS_2, the mechanism of dielectric loss is necessary to be explored. Polarization loss and conductivity loss are the primary dielectric losses, according to Debye theory [59]. Moreover, interface polarization and dipolar polarization can be used to categorize polarization loss [60]. The Cole-Cole semicircle can be used to evaluate this dielectric loss, which is calculated as:

$$\left(\varepsilon' - \frac{\varepsilon_s - \varepsilon_\infty}{2}\right)^2 + (\varepsilon'')^2 = \left(\frac{\varepsilon_s - \varepsilon_\infty}{2}\right)^2 \quad (7)$$

The Cole-Cole curve usually be consisted of multiple semicircles, suggesting the presence of a polarization loss process [61]. Each semicircle represents a relaxation process, and the larger radius represents the stronger relaxation loss [62]. As shown in Fig. 8, several semicircles exist in the Cole-Cole curves of all the samples, indicating the presence of multiple polarization relaxation [63]. Compared with the pure MoS_2, the larger semicircle of SiO_2@MoS_2 is mainly caused by the formation of core-shell structure. Plenty of heterogeneous interfaces (between MoS_2 and SiO_2) in SiO_2@MoS_2 nanospheres greatly promote interface polarization. Due to their different dielectric properties and conductivity, the accumulated charges caused by the Schottky barrier were formed at the heterogeneous interface and interacted with the incident EM wave to promote dielectric loss [64]. Besides, the 1T-MoS_2 also promotes the improvement of conductivity loss. There are a large number of defects (such as S vacancies and Mo vacancies) and binding charges in the MoS_2 nanosheets, which can act as dipoles to enhance dipole polarization [65,66]. The core-shell structure is conducive to the formation of interconnected three-dimensional conductive networks. When the EM wave enters the materials, electrons would absorb energy and the EM wave will be dissipated by the generated micro-current move in conductive networks [67] (see Fig. 9).

In order to further study the multiple polarization relaxation of the materials, electron holographic technology characterization is performed. The off-axis holography results show that a large number of positive and negative charges accumulate around the interfaces between MoS_2/MoS_2 nanosheets, which are confirmed by the alternating blue, red and yellow regions[68]. Under high-frequency EM fields, positive and negative charges can accumulate around the interface of MoS_2 nanosheets, which causes a significant loss of interface polarization [69,70]. In SiO_2@ MoS_2-2, a large number of interfaces between ultrathin MoS_2 nanosheets provide large-scale interface polarization points. Due to the electromagnetic field, the symmetry center of the positive and negative charges will gradually deviate, creating a vast dipole polarization [71]. As a result, the buildup of charges is crucial in the attenuation of EM waves.

The efficacy of EM wave absorption is significantly influenced by two key factors: impedance matching and attenuation constant (α) [72,73].

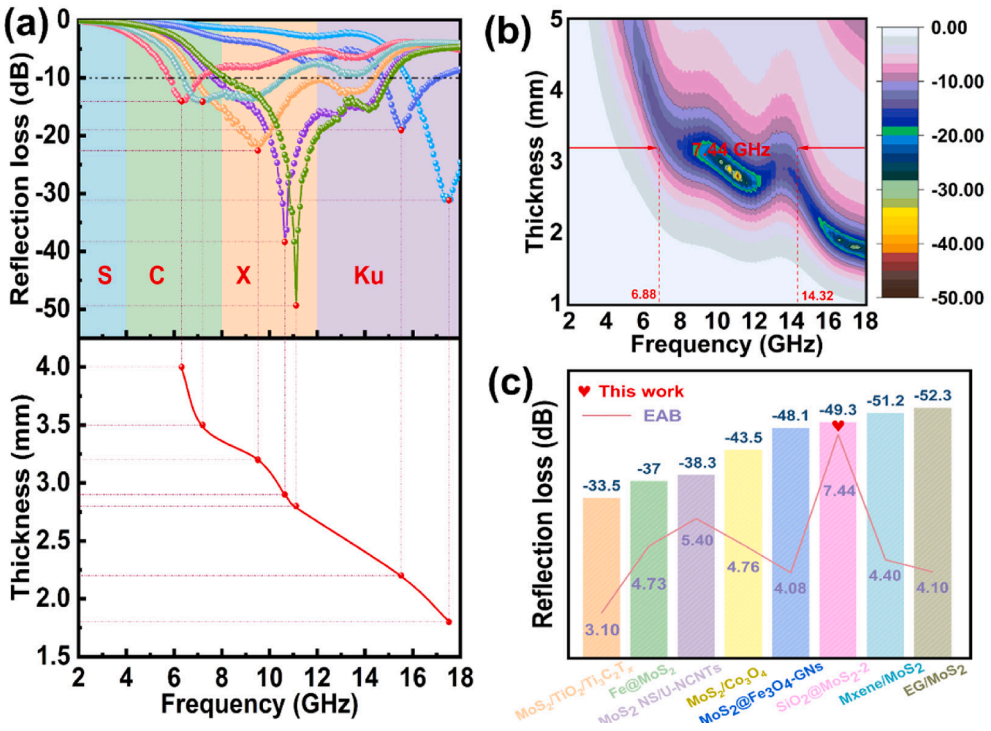


Fig. 7. (a) EM wave absorption properties and the corresponding thickness simulation of SiO_2@MoS_2-2; (b) Contour map of the values of RL of SiO_2@MoS_2-2; (c) A comparison of various MoS_2-based composites for EMW absorption.

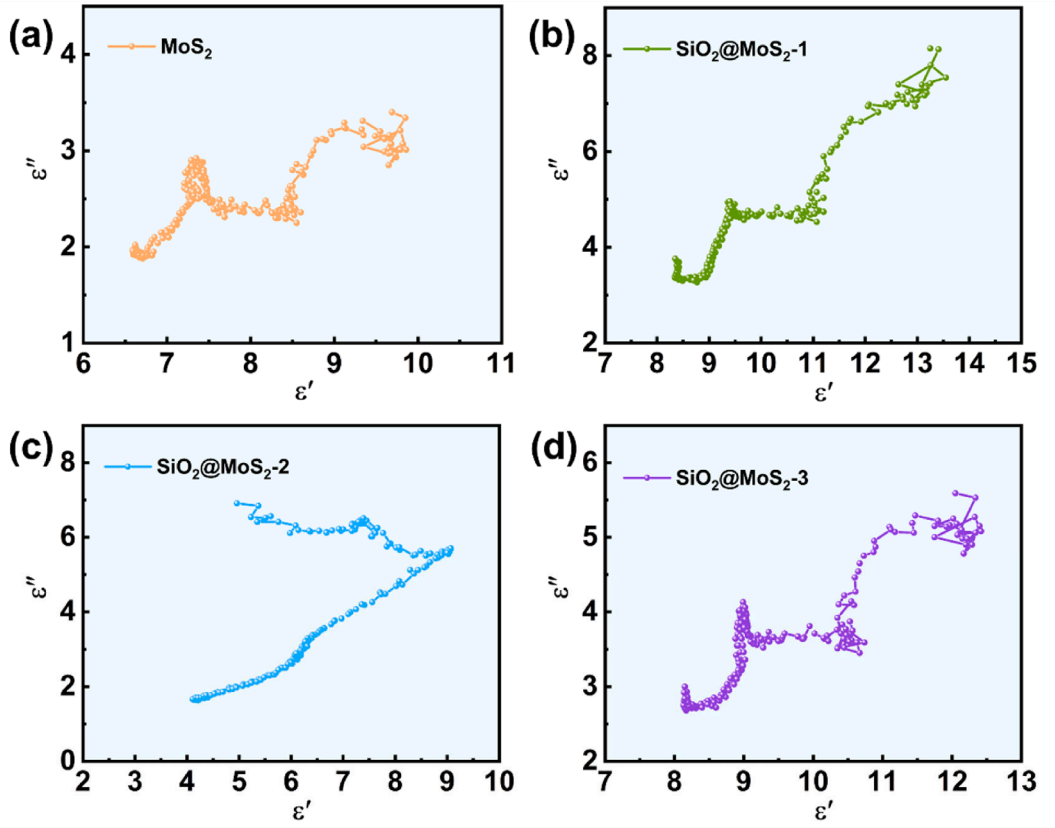


Fig. 8. Cole-Cole curves of (a) MoS_2; (b) SiO_2@MoS_2-1; (c) SiO_2@MoS_2-2; (d) SiO_2@MoS_2-3.

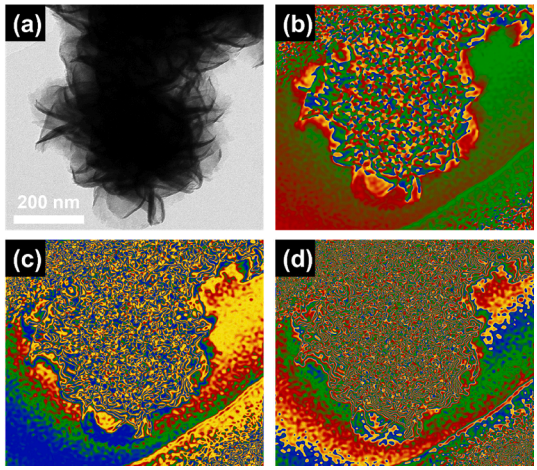


Fig. 9. (a) TEM images; (b-d) the corresponding off-axis electron holograms of the spatial potential and charge density of SiO_2@MoS_2-2.

The α can be analyzed using transmission line theory:

$$Z = \left|\frac{Z_{in}}{Z_0}\right| = \sqrt{\left|\frac{\mu_r}{\varepsilon_r}\right|}\tanh\left(j\frac{2\pi fd}{c}\sqrt{\mu_r\varepsilon_r}\right) \tag{8}$$

$$\alpha = \frac{\sqrt{2}\pi f}{c} \times \sqrt{(\mu''\varepsilon'' - \mu'\varepsilon') + \sqrt{(\mu''\varepsilon'' - \mu'\varepsilon')^2 + (\mu''\varepsilon'' + \mu''\varepsilon')^2}} \tag{9}$$

When the value of $|Z_{in}/Z_0|$ is near to 1, more EM waves can be absorbed [74,75]. The two-dimensional contour maps of the $|Z_{in}/Z_0|$ values of the materials are shown in Fig. 10. Compared with pure MoS_2, $|Z_{in}/Z_0|$ closed to 1 for SiO_2@MoS_2-2 has larger area, indicating that the proper introduction of SiO_2 can optimize impedance matching [76]. Secondly, the attenuation ability of materials for incident EM waves can be described by α value, and the larger α value represents the stronger attenuation ability [77–79]. As shown in Fig. 5d, the α values of all samples increase gradually with increasing frequency. All SiO_2@MoS_2 samples have higher α values compared to pure MoS_2, indicating that the core-shell structure improves impedance matching and EM wave attenuation. The α value of SiO_2@MoS_2-2 is not the largest, but more EM waves can penetrate the materials because of its better impedance matching. SiO_2@MoS_2 possesses the characteristics of thin (2.8 mm), wide (7.44 GHz), and strong (−49.35 dB), exhibiting excellent EM wave absorption ability.

Fig. 11 displays the EM wave absorption mechanism diagram of the core-shell structured SiO_2@MoS_2 nanospheres. First, the ultra-thin MoS_2 nanosheets are conducive to achieving attenuation effect through the multiple reflection and scattering of EM waves. Second, SiO_2@MoS_2 develops conductive networks, which act as a pathway for electron migration to promote conduction loss. Third, a significant number of interfaces are created in SiO_2@MoS_2, causing strong interfacial polarization under EM fields. In addition, the dipole polarization caused by abundant defects and accumulated charges at interface can effectively dissipate EM waves. Therefore, the core-shell structured SiO_2@MoS_2 exhibits excellent EM wave absorption capacity due to proper impedance matching and multiple relaxation loss.

4. Conclusions

In summary, the core-shell structured SiO_2@MoS_2 nanospheres are successfully fabricated using a simple hydrothermal process. The results show that SiO_2@MoS_2 owns the RL_{min} of − 49.35 dB at 2.8 mm, and its EAB is 7.44 GHz at 3.2 mm. The core-shell structured SiO_2@MoS_2 nanospheres optimize impedance matching and elevate the attenuation effect, which markedly enhances the EM wave absorption performance.

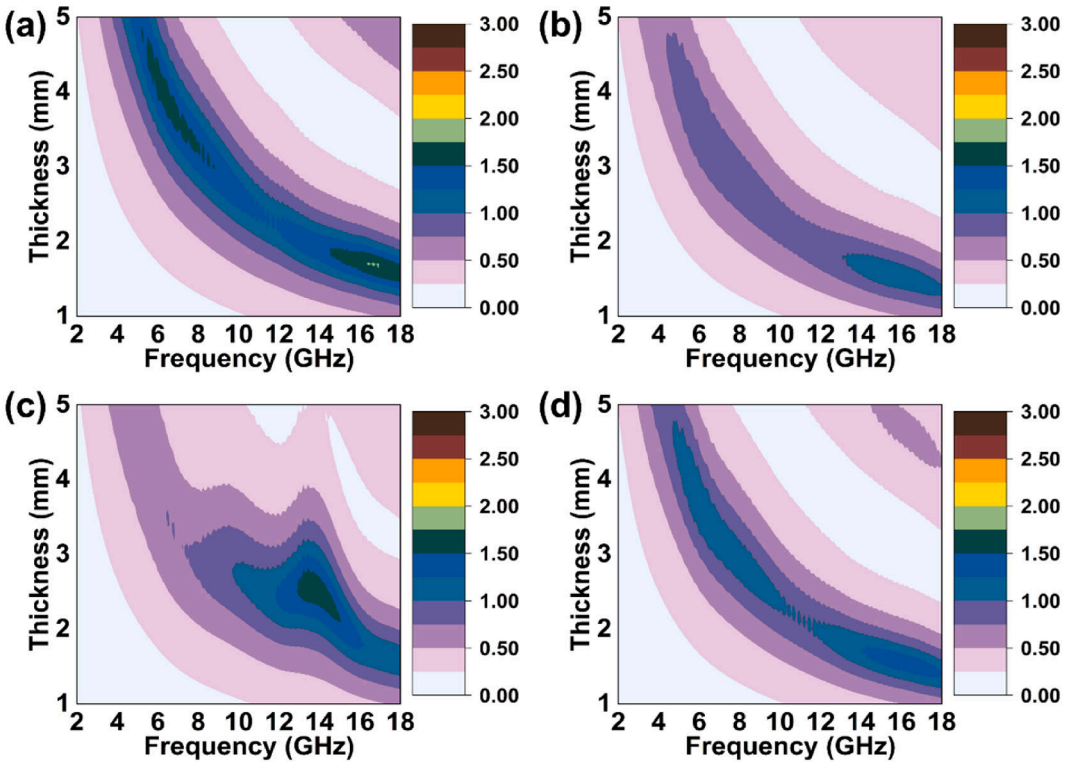


Fig. 10. Contour map of the values of Z_{in} of (a) MoS_2; (b) SiO_2@MoS_2-1; (c) SiO_2@MoS_2-2; (d) SiO_2@MoS_2-3.

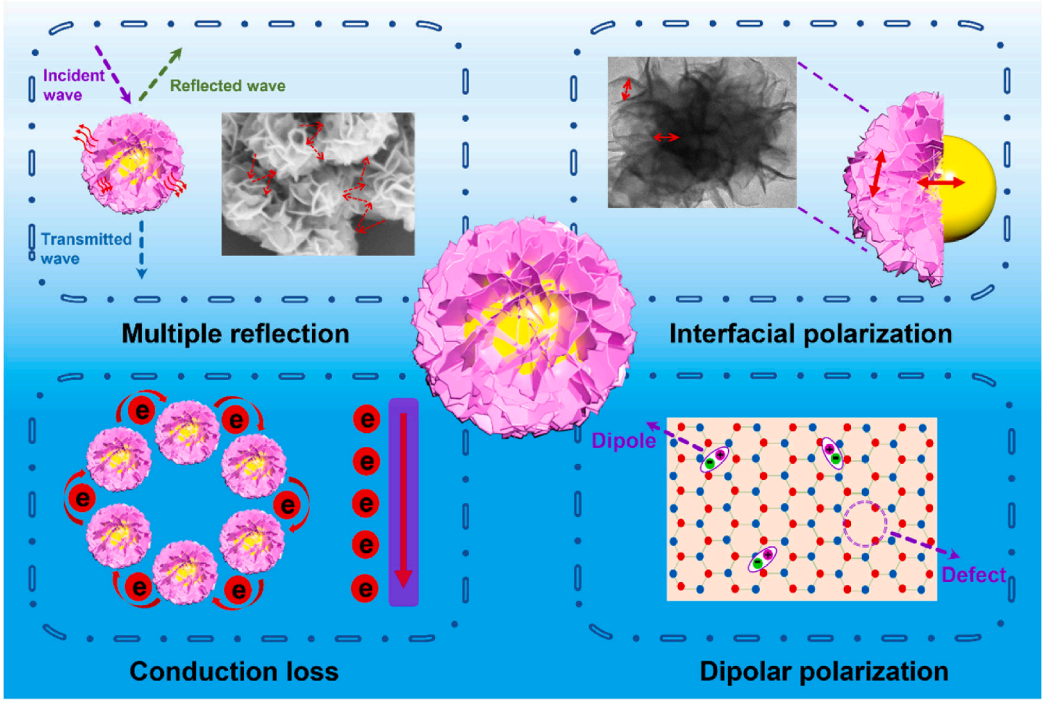


Fig. 11. Schematic illustration of the EM wave absorption mechanism for SiO_2@MoS_2 nanospheres.

The multiple loss mechanisms including interfacial polarization, dipole polarization, conduction loss and multiple scattering and reflection contribute that the majority of EM waves can be attenuated. In summary, this work offers a fresh option for the synthesis of high-efficiency EM wave-absorbing MoS_2-based absorbers.

CRediT authorship contribution statement

Xuewen Jiang: Conceptualization, Validation, Investigation, Resources, Data curation, Writing – original draft. **Huanhuan Niu:** Data curation, Visualization. **Jinlu Li:** Data curation, Investigation. **Mengru Li:** Data curation, Investigation. **Chao Ma:** Writing – review & editing, Visualization, Supervision. **Rui Zhang:** Writing – review & editing.

Hailong Wang: Writing – review & editing. **Hongxia Lu:** Writing – review & editing. **Hongliang Xu:** Writing – review & editing. **Bingbing Fan:** Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Data availability

Data will be made available on request.

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